# Do Now · retrieval

## Compiler stages, linkers, loaders and libraries

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which program translates assembly mnemonics?

### Q2

What does a compiler typically produce?

### Q3

What is a runtime interpreter?

### Q4

Simplify ¬(A∨B) with De Morgan's law, then evaluate A=0,B=1.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Understand — Lexical analysis identifies tokens, while syntax analysis checks grammar and generates a parse structure.

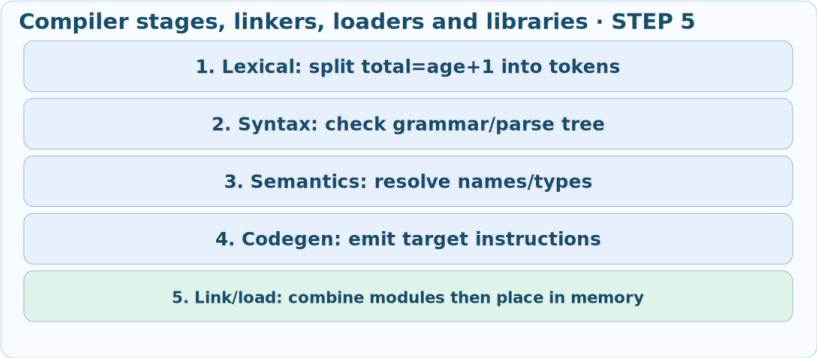
## Core idea 2

Apply — Semantic analysis checks rules beyond grammar such as type compatibility or declared identifiers; optimisations can transform intermediate code.

## Core idea 3

Evaluate — A linker combines object files and resolves symbols/libraries; a loader places the executable into memory for execution.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

lexer — Recognises source tokens.

parser — Checks grammatical structure.

semantic analysis — Checks program meaning and types.

linker — Resolves symbols across modules.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

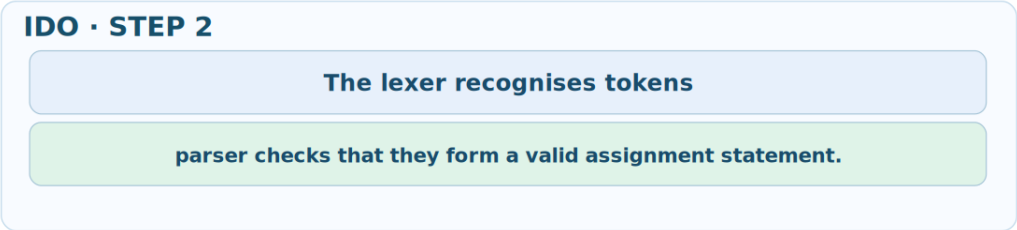
## Problem

Analyse `total = count + 1;` into tokens and syntax structure.

## Follow the worked method

1. Predict first — The lexer recognises tokens
2. Model answer — parser checks that they form a valid assignment statement.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
The linker finds its definition in the module/library and fixes up the reference before execution.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Distinguish lexical, syntax and semantic analysis. [6]

## Question 2

Describe the roles of code generation, linking and loading. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

A source file has valid syntax but calls undefined MethodX(). Explain which analyses or stages may detect failure, and why. [6]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

A parser guarantees that every variable in a syntactically valid program was declared.

## Use key vocabulary from memory

1. Explain lexer in your own words.

2. Explain parser in your own words.

3. Explain semantic analysis in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.